

constantly causing it to change direction from following a straight line (the “natural” path of any moving object) to following a curve—known as its orbit around the Sun.

The third Law of Motion is the one that causes people the most trouble. Newton used the word “action” to mean “force” when he said that for every action there is an equal and opposite reaction. For example, when I fire a rifle, the action pushes the bullet out of the barrel and the reaction makes the rifle butt kick against my shoulder. If I stood on an ice-skating rink and threw a heavy medicine ball away from me, the reaction would make me slide backward on the ice. When a spacecraft fires its rear engine in space, exhaust gases are pushed out of the vents, producing a reaction that makes the rocket accelerate forward. It is this law that explains how atoms bounce off each other, what happens when pool balls collide, and why head-on car collisions can be even worse than running into a brick wall, because the combined speed of the two cars has to be taken into account.

a momentous collision

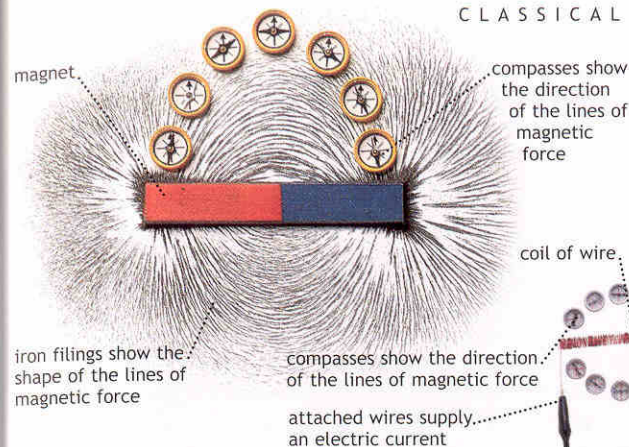
If a moving car hits a brick wall, the scattering of the bricks carries some of the momentum away. But in a head-on collision, their combined momentum is completely absorbed by the cars, causing much more damage.



Maxwell's equations

Until the end of the 1800s, Newton's three laws seemed to define the material world completely. Even the behavior of atoms and the newly discovered electron could be explained by a combination of Newton's laws and the application of electric and magnetic forces.

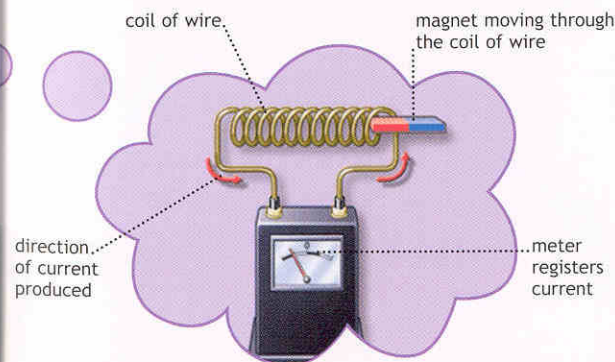
Maxwell provided an explanation of these two forces, the other half of the physical world, which seemed to complete the picture. Maxwell built on the research by Michael Faraday into electricity and magnetism. During the first half of the 1800s, Faraday



fields of force

The lines of magnetic force making up the field around a bar magnet can be traced using iron filings, or by placing magnetic compasses near the magnet. The compass needles line up along the field lines. When an electric current flows through a coil of wire, it causes a similar field of magnetic force to form, but only while the electric current is flowing.

developed the key concepts of lines of force, and magnetic and electric “fields.” He invented the electric motor and the dynamo, and discovered that a changing magnetic field always produces an electric field, and that changing electric field always produces a magnetic field. A field can be thought of as the region over which the force involved has influence. It can be visualized by what happens when a bar magnet is placed under a sheet of paper covered with iron filings and the paper is gently tapped. The filings align themselves in curving patterns around the magnet, marking out the shape of its field. Each of the curved lines linking the south pole of the magnet to its north pole is a line of force.



dynamic physics

A dynamo, or electric generator, works because a magnet moving past a wire makes an electric current flow in the wire. Michael Faraday discovered that pushing a magnet into a coil of wire connected to a meter to measure electric current caused the needle of the meter to flicker.